

RAKSHITHA H P

Data Science Undergraduate | Software Engineering | Python | SQL | Backend Development

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PROFESSIONAL SUMMARY

Data Science undergraduate with hands-on software development experience in Python, SQL, backend development, REST APIs, machine learning, data processing, and AI systems. Built a fraud detection microservice, retrieval augmented generation pipeline, and MRI classification application. Seeking a Software Engineering internship focused on Python, backend development, APIs, databases, and production software.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, Node.js

Backend Development: Flask, Node.js, Express, REST APIs, object-oriented programming

Machine Learning and AI: Scikit-Learn, TensorFlow, Keras, CNN, VGG16, ResNet50, LangChain, RAG

Data and Databases: MySQL, SQLite, ChromaDB, Redis, data cleaning, data validation, analytical reporting

Tools: Streamlit, Power BI, Tableau, Excel, Google Sheets, GitHub, GitHub Copilot, ChatGPT

EXPERIENCE

Python Intern | EmbeddedFru | Remote

July 2023 - January 2024

- Developed Python applications for data processing and workflow automation, applying object-oriented programming principles to build modular and maintainable pipelines.
- Executed SQL queries to extract, filter, and aggregate records from relational databases, supporting reporting and business insight generation.
- Organized and processed business datasets using Python, applying data cleaning and validation to produce structured, analysis-ready outputs.

PROJECTS

SIM Swap Attack Detection | SecurBank Fraud Analytics

Python, Flask, Scikit-learn, SQLite

- Built a real-time fraud detection microservice and analyst dashboard using a hybrid machine learning and heuristic risk engine.
- Implemented a step-up OTP verification flow and telecom webhook endpoint using REST API design and SQL-backed data modeling for users, transactions, and OTP tokens.

Nyaya Sathi | AI Legal Assistant | Major Project

Python, LangChain, RAG, ChromaDB, Redis, Streamlit

- Building the AI backend for a legal guidance assistant that answers user queries from Indian legal sources with traceable citations.
- Implemented the retrieval augmented generation pipeline end to end, including document ingestion, embeddings, vector search, prompt orchestration, and Redis-backed query and chat caching.

Brain Tumor Detection | Brain MRI Classification

Python, TensorFlow, Keras, CNN, VGG16, ResNet50

- Trained and compared a custom CNN, VGG16 with transfer learning, and ResNet50 fine-tuned for binary and multi-class MRI classification, achieving up to 99.07% binary and 96.18% multi-class accuracy.
- Built an interactive Streamlit application for image upload and model comparison.

EDUCATION

B.E. in Data Science

MVJ College of Engineering, Bangalore

August 2024 - June 2027

4th Year | CGPA: 6.7 | Coursework: Artificial Intelligence and Machine Learning, Data Structures, DBMS, Data Visualization, Big Data Analytics

Diploma in Computer Science and IT

Siddaganga Polytechnic, Tumkur

July 2021 - June 2024

CGPA: 8.88 | Coursework: Basics of Programming, Networking, Full Stack Development

LEADERSHIP AND ACTIVITIES

- Participated in 3 or more hackathons at Maharaja Institute of Technology, Sapthagiri NPS University, and Project Expo at MVJ College of Engineering.
- Department Co-Ordinator, Vertechx Event — managed end-to-end event logistics with strong organizational discipline.
- Web Solutions Strategist, Toastmasters International District 92 — developed communication and leadership skills.